Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors

Weekly Progress Report — Week 9 (Final Milestone)

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1 Introduction

Week 9 marks the completion of the Fabrix project, addressing our mid-semester commitment to develop a **multi-class physical defect taxonomy** and perform the **end-to-end integration and benchmark validation** of our classical DIP inspection pipeline. While our earlier work focused on binary screening (clean versus defective), automated manufacturing requires diagnosing the exact physical failure etiology so loom operators can take targeted corrective actions. In Week 9, we engineered an interpretable, deterministic classification engine that maps all 12 benchmark defect codes into four physical failure families without using machine learning.

2 Invariant Shape Characterization via Moments

To recognize defect geometries independently of conveyor translation, scale, or orientation, we implemented classical invariant moment descriptors:

- **Translation and Scale Invariance:** Centroid-referenced central moments achieve position invariance, while area normalization eliminates sensitivity to optical magnification and defect size.
- **Hu Invariant Moments:** The first invariant moment captures geometric compactness, separating circular flaws from elongated streaks. Higher-order moments describe eccentricity and shape asymmetry.
- **Inertia Tensors and Principal Orientation:** Diagonalizing the second-order inertia matrix yields the flaw's principal axis angle and aspect ratio, cleanly distinguishing longitudinal warp breaks from transverse weft disruptions.

3 Resolving Hole vs. Stain Ambiguity via Laws' Texture Energy

A key challenge highlighted in our mid-semester review was separating micro-punctures (holes) from surface oil spots. In grayscale imagery, both appear as dark, compact circular spots with nearly identical aspect ratios, rendering bounding-box geometry ineffective. We solved this strictly within classical DIP:

- **Kenneth Laws' Spot Energy (S_5S_5):** We convolve flaw candidates with Laws' symmetric spot kernel to measure the presence of underlying woven yarn crossings.
- **Texture Intactness Ratio:** For physical holes, yarns are severed or missing, causing texture energy to collapse to zero. For photometric oil stains, the woven yarns remain physically intact beneath the liquid, preserving strong texture energy.
- **Boundary Acutance:** Flaw perimeter gradients confirm the diagnosis: mechanical tears exhibit sharp edges, whereas liquid stains show diffuse capillary penumbras.

4 Deterministic Taxonomy and System Integration

Using these descriptors, we structured a hierarchical deterministic decision tree that classifies all 12 AITEX codes into four actionable failure classes:

1. **Warp Flaws (Longitudinal):** High aspect ratio and vertical orientation (broken ends, creases, knots).
2. **Weft Flaws (Transverse):** Low aspect ratio and horizontal orientation (broken picks, weft curling, cracks).
3. **Micro-Punctures (Voids):** Compact shape, collapsed texture energy, and razor-sharp edges (holes).
4. **Photometric Stains (Contaminants):** Compact shape, intact texture energy, and diffuse penumbra (oil spots).

All modules were integrated into an end-to-end pipeline and validated on the 81-roll test split across seven commercial weaves. Our specialized descriptors rescued high-clutter fabrics: 2D-FFT Butterworth notch filtering eliminated chevron patterns on herringbone twills, and Radon projection profiling suppressed vertical ribs on striped fabrics to achieve zero false loom halts.

5 Work Completed This Week

- Implemented Kenneth Laws' Texture Energy Measures (S_5S_5) and boundary acutance profiling to resolve the hole versus stain ambiguity.
- Built the deterministic 4-class taxonomy decision tree mapping all 12 AITEX defect codes.
- Integrated preprocessing, Fourier notch filtering, adaptive morphology, and taxonomy into the master pipeline.
- Completed full benchmark evaluation and threshold sensitivity sweeps across all 81 test rolls.
- Expanded and verified the unit test suite to 73 tests (all passing cleanly).

6 Conclusion

Week 9 completes the goals of the Fabrix project. By combining 2D Fourier filtering, Radon alignment, weave-adaptive morphology, and Laws' Texture Energy, our framework delivers a fully interpretable, real-time fabric defect screening and multi-class diagnosis pipeline using strictly classical Digital Image Processing.

End of Week 9 Report